

2026

Physics

Full Marks: 50

Pass Marks: 17

Time: 2-hours

(The figures in the margin indicate full marks for the questions)

Allotment of Marks

Q. No. 1 carries 1 mark each	$1 \times 6 = 6$
Q. No. 2 carries 2 marks each	$2 \times 8 = 16$
Q. No. 3 carries 3 marks each	$3 \times 6 = 18$
Q. No. 4 carries 5 marks each	$5 \times 2 = 10$
Total Marks = 50	

1. Answer the following questions:

$1 \times 6 = 6$

(a) The SI unit of electric potential is—

- (i) Volt (ii) Coulomb
(iii) Joule (iv) Newton

(b) The work done in moving a charge between two points on the same equipotential surface is—

- (i) Positive (ii) Negative
(iii) Zero (iv) Infinite

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(c) The SI unit of resistivity is—

- (i) $\Omega \text{ m}$ (ii) Ω/m
(iii) $\Omega \text{ m}^2$ (iv) $\Omega^2 \text{ m}$

(d) The magnetic field at the centre of a circular current-carrying coil is directed along its—

- (i) Radius (ii) Diameter
(iii) Axis (iv) Circumference

(e) The magnetic susceptibility of a diamagnetic substance is—

- (i) Positive (ii) Negative
(iii) Zero (iv) Infinite

(f) According to Faraday's law, induced emf is produced due to the change in—

- (i) Electric field (ii) Current
(iii) Magnetic flux (iv) Resistance

2. Answer any eight questions from the following:

$2 \times 8 = 16$

(a) Using Gauss's law, explain why the electric field inside a hollow conducting sphere is zero.

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- (b) Define electric dipole moment. Write its SI unit.
- (c) Define current density. Write its SI unit.
- (d) A wire carries a current of 5 A. If its cross-sectional area is $2 \times 10^{-6} \text{ m}^2$, calculate the current density.
- (e) Write any two properties of magnetic field lines.
- (f) State Ampere's circuital law and mention one application of the law.
- (g) Write the Coulomb's law in vector form. What is unit of permittivity of free space?
- (h) State Faraday's first law of electromagnetic induction.
- (i) What is self-induction?

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3. Answer all questions from the following:

$$3 \times 6 = 18$$

- (a) State Gauss's theorem.

Hence obtain the expression for the electric field due to an infinitely long straight line charge.

$$1 + 2 = 3$$

Or /

A charge of $5\mu\text{C}$ is placed at the centre of a cube of side 20 cm. Calculate the electric flux through each face of the cube.

$$1 + 2 = 3$$

- (b) Write the law that is applied in wheatstone bridge principle.

Draw a wheatstone bridge.

$$1\frac{1}{2} + 1\frac{1}{2} = 3$$

Or /

Derive the relation between current density and drift velocity.

3

- (c) Using Biot-Savart law, obtain the expression for the magnetic field at the centre of a circular current-carrying loop.

3

Or /

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A circular coil of radius 10 cm carries a current of 5 A.

Calculate the magnetic field at its centre.

1 + 2 = 3

- (d) Differentiate between diamagnetic and paramagnetic substances. Give one example of each.

2 + 1 = 3

Or /

Define magnetic susceptibility. Establish the relation $\mu^r = 1 + \chi_m$.

- (e) State Faraday's laws of electromagnetic induction:

A coil of 250 turns experiences a change in magnetic flux from 0.08 Wb to 0.02 Wb in 0.02 s. Calculate the induced emf.

1 + 2 = 3

Or /

State Lenz's law. Explain how it is consistent with the law of conservation of energy.

1 + 2 = 3

- (f) Define self-inductance. Obtain the expression for the energy stored in an inductor.

1 + 2 = 3

Or /

A coil of self-inductance 4 H carries a current of 3 A. Calculate the energy stored in the magnetic field.

1 + 2 = 3

Turn Over

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4. Answer the following questions: (Any two)

$$5 \times 2 = 10$$

(a) Using Gauss's law, derive the expression for the electric field due to a uniformly charged thin spherical shell at

(i) an external point.

(ii) an internal point.

Draw the variation of electric field with distance from the centre.

$$2 + 2 + 1 = 5$$

Or /

A battery of emf 12 V and internal resistance $1 \, \Omega$ is connected to an external resistance of $5 \, \Omega$. Calculate the current in the circuit and the terminal potential difference of the battery.

$$2 + 2 + 1 = 5$$

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- (b) State Faraday's laws of electromagnetic induction. Explain Lenz's law and show that it is consistent with the law of conservation of energy.

2 + 2 + 1 = 5

Or /

A rectangular coil of 500 turns having an area of $4 \times 10^{-3} \text{ m}^2$ rotates with a uniform angular speed of 120 rad s^{-1} in a uniform magnetic field of 0.5 T.

- (i) Derive the expression for the induced emf.
(ii) Calculate the maximum induced emf.

3 + 2 = 5

- (c) For the above network figure— (i) Determine the equivalent capacitance and (ii) The charges in each capacitor.

DIAGRAM PLACEHOLDER

A rectangular capacitor network. The four capacitors are $C_1 = 10 \text{ } \mu\text{F}$ on the left side, $C_2 = 10 \text{ } \mu\text{F}$ on the top side, $C_3 = 10 \text{ } \mu\text{F}$ on the right side, and $C_4 = 10 \text{ } \mu\text{F}$ on the bottom side. A 500 V potential difference is applied between the two bottom nodes. The top branch contains C_2 ; the left and right branches contain C_1 and C_3 ; C_4 connects the two lower nodes.

2 + 3 = 5

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Or /

DIAGRAM PLACEHOLDER

A resistor network from A to C. R_1 is in series from A to junction B. From B to C there are two parallel branches: the upper branch contains R_2 carrying current I_1 , and the lower branch contains R_3 carrying current I_2 . The total current I flows through R_1 and then from the parallel combination to C. A voltage V is indicated across the network from A to C.

For the above network show that,

$$I = \frac{V(R_2 + R_3)}{R_1R_2 + R_1R_3 + R_2R_3}$$